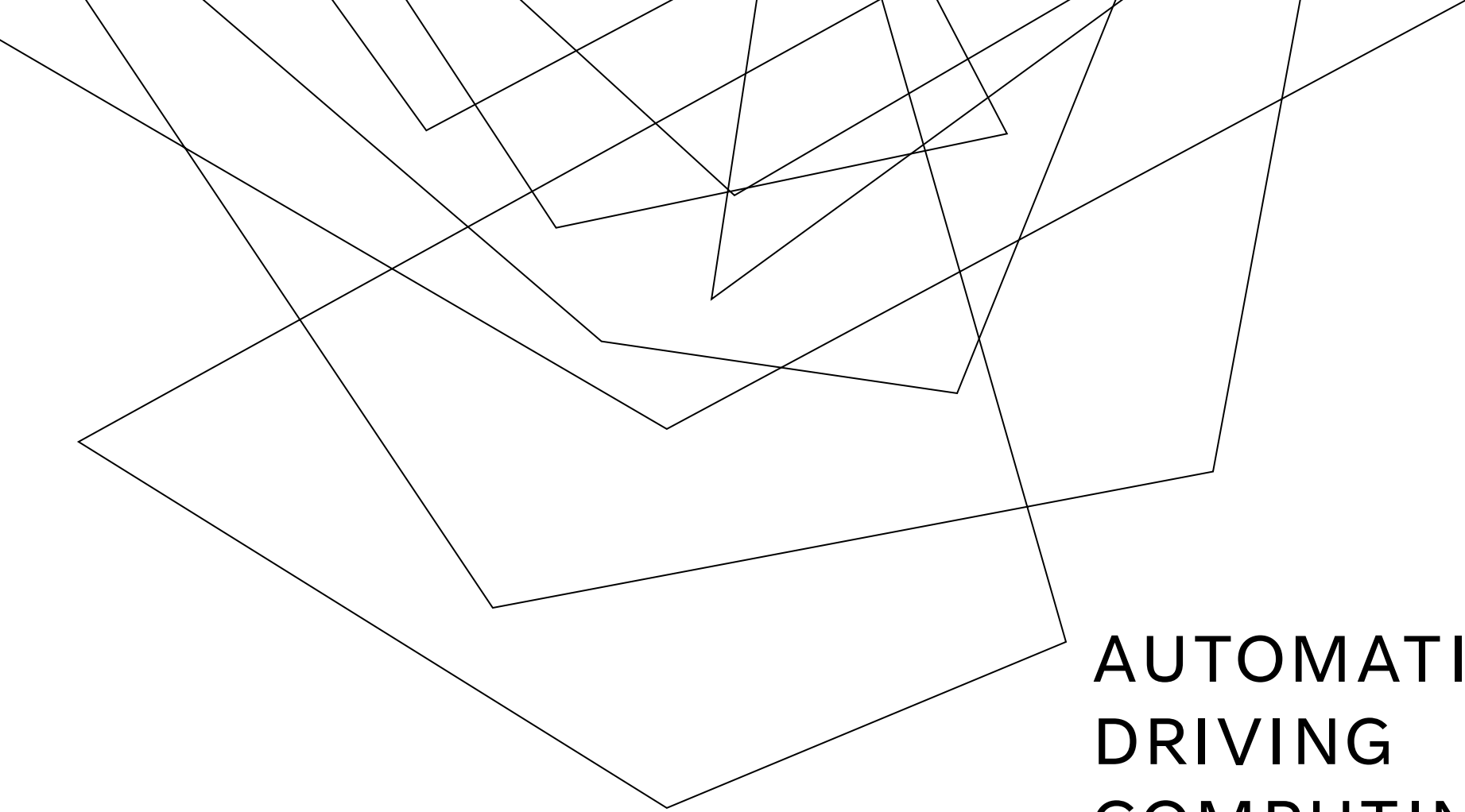


A series of thin, black, overlapping lines forming various geometric shapes like triangles and polygons, creating a complex, abstract pattern in the upper left portion of the slide.

AUTOMATIC DRIVING COMPUTING

2. SENSORS





목차

목차

1. Role of sensors in autonomous driving cars
2. Types of sensors in autonomous driving cars
3. Case study



An abstract graphic design featuring two thin, dark grey lines that intersect on a light grey background. One line runs diagonally from the top-left towards the bottom-right, while the other runs from the top-right towards the bottom-left. The intersection point is located to the left of the text.

ROLE OF SENSORS IN AUTONOMOUS DRIVING CARS

SENSORS

자율주행차량에서 운전자의 “눈” 역할



SENSORS

감각기관별 정보수용량 비교

시각	83%
청각	11%
후각	3%
촉각	2%
미각	1%



TECH STACKS OF AUTONOMOUS DRIVING

Autonomy Stack - Sensors



An abstract graphic design featuring two thin, dark grey lines that intersect on a light grey background. One line runs diagonally from the top-left towards the bottom-right, while the other runs from the top-right towards the bottom-left. The intersection point is located to the left of the text.

TYPES OF SENSORS IN AUTONOMOUS DRIVING CARS

CAMERA

(디지털) 카메라의 원리

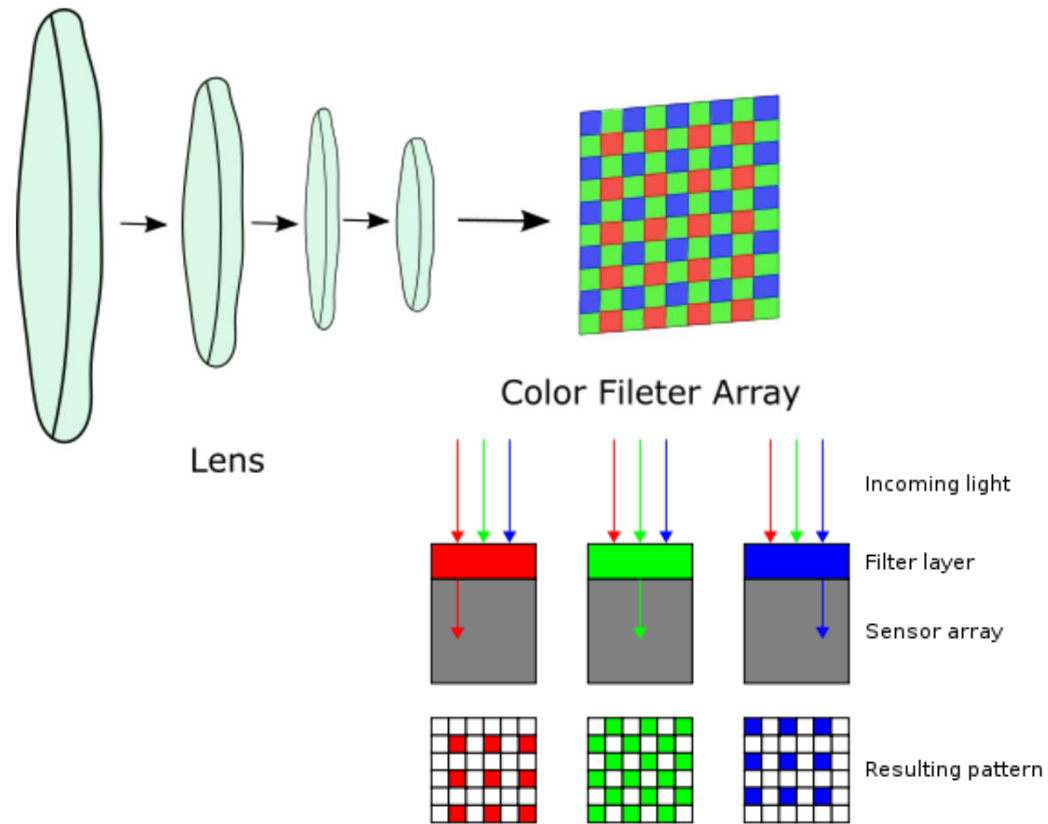
1) 빛 (Photon)이 물체에서 반사됨



CAMERA

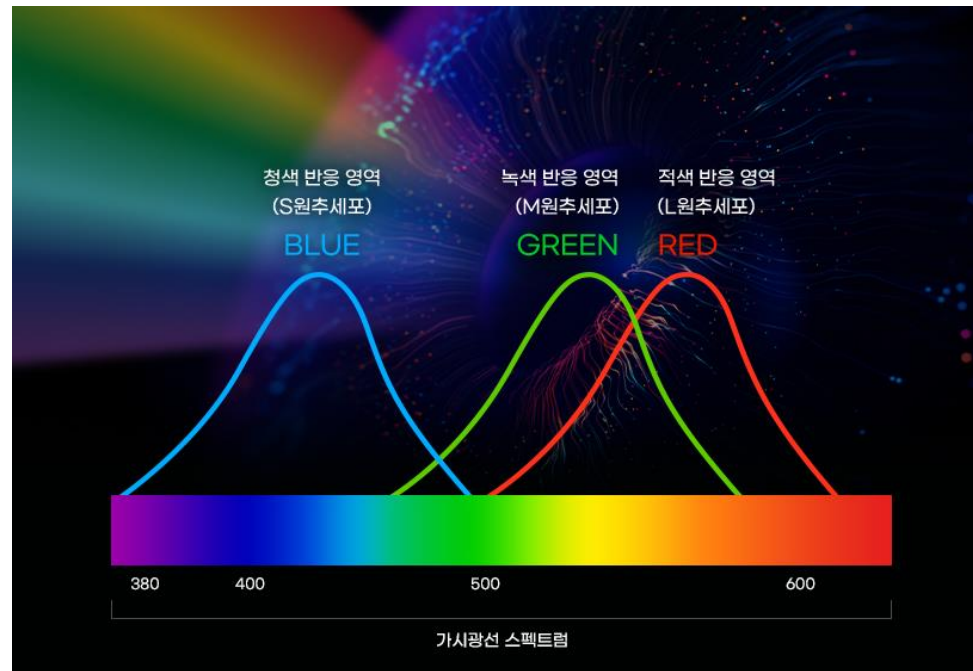
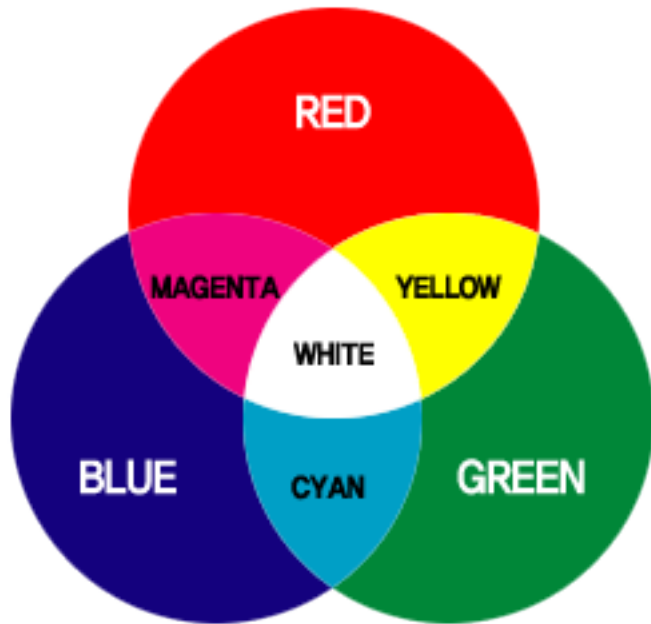
(디지털) 카메라의 원리

2) 빛이 격자 (Array) 모양으로 배열된 RGB 필터를 거침



CAMERA

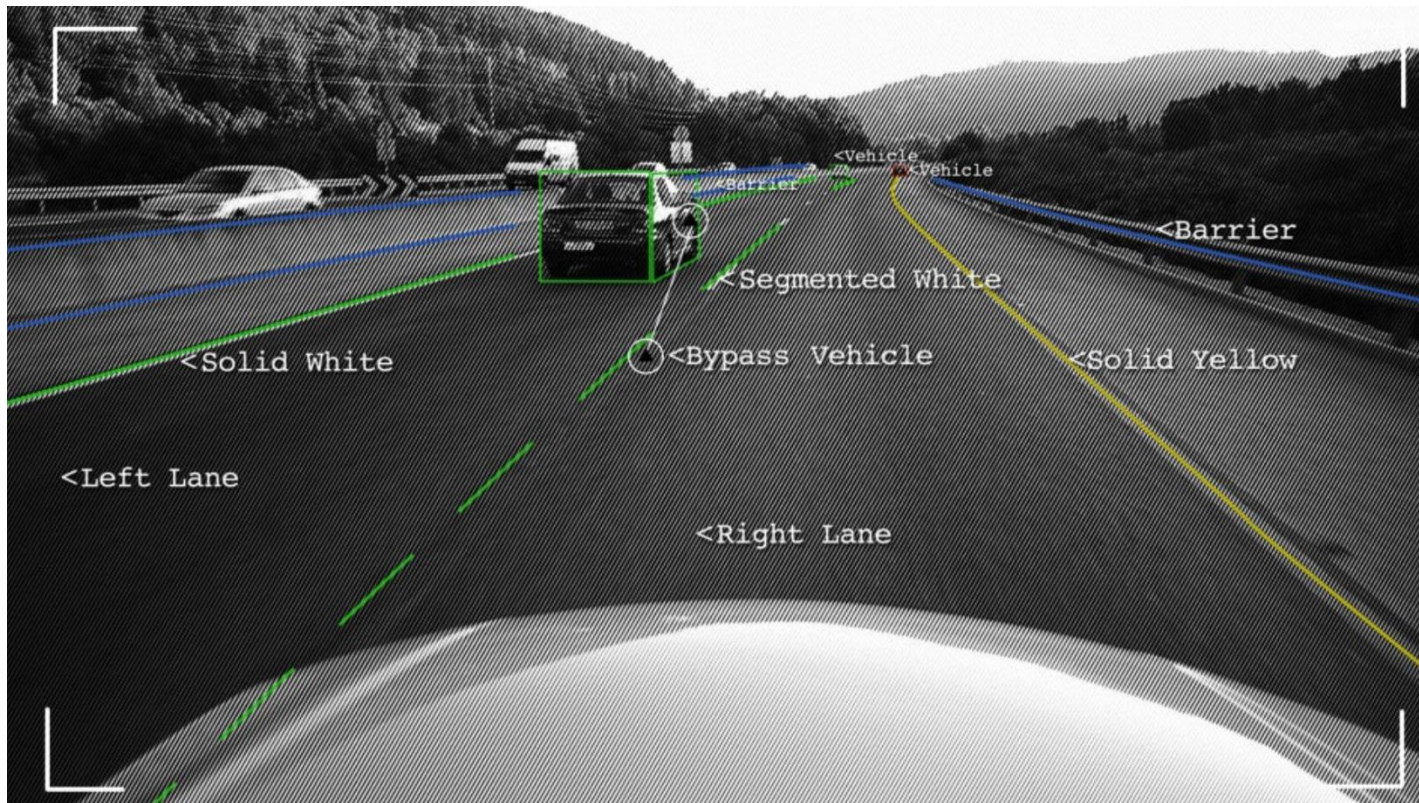
cf) 빛의 3원색



CAMERA

장점

- 화질이 좋음
- 색 감지 가능



CAMERA

장점

- 화질이 좋음
- 색 감지 가능

단점

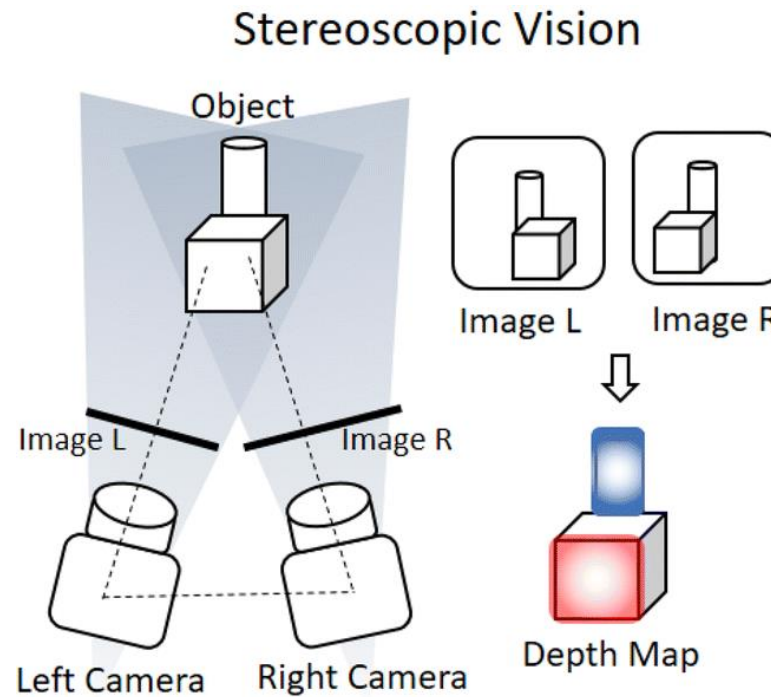
- 감지 각도가 제한적
- 깊이 (Depth) 정보를 알 수 없음
- 날씨 등 외부 환경에 영향
- 밤에는 정보 취득이 불가능



CAMERA

Cf) Stereo camera

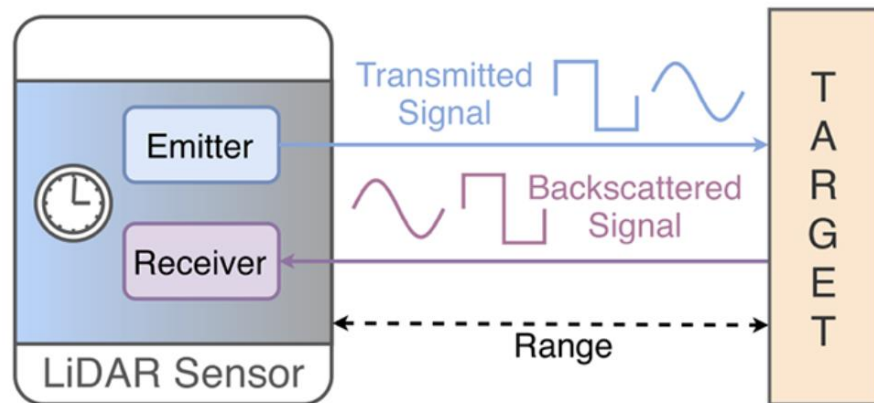
두 개 이상의 카메라를 배치하여 동일한 장면을 촬영
➔ 깊이 (Depth) 정보 계산 가능



LIDAR

LiDAR (Light Detection and Ranging)

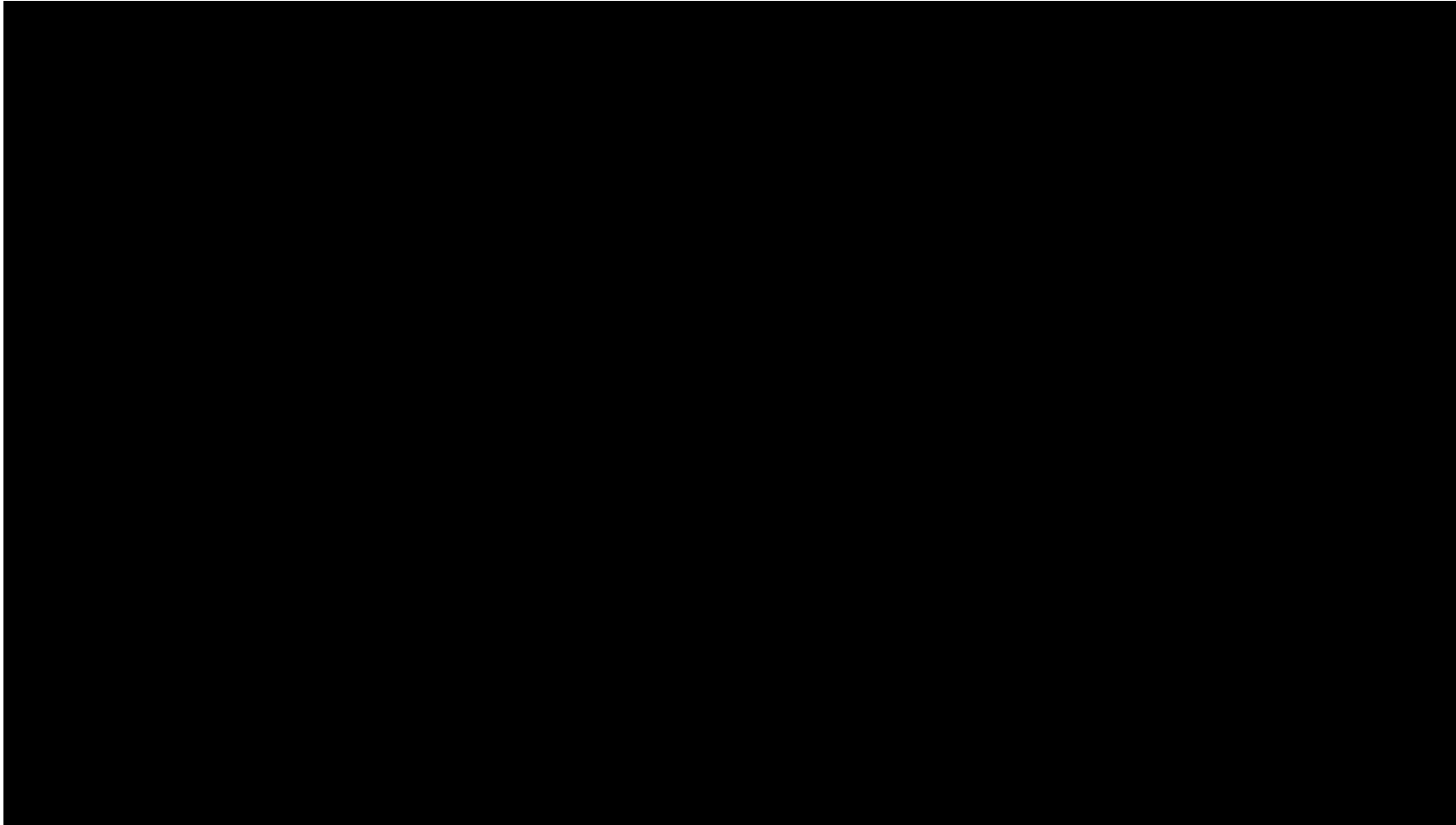
레이저 신호를 발사하고 그것이 물체에 맞고 반사되어 되돌아 오는 전자파를 분석하여 대상물과의 거리를 측정



Source: <http://velodynelidar.com>



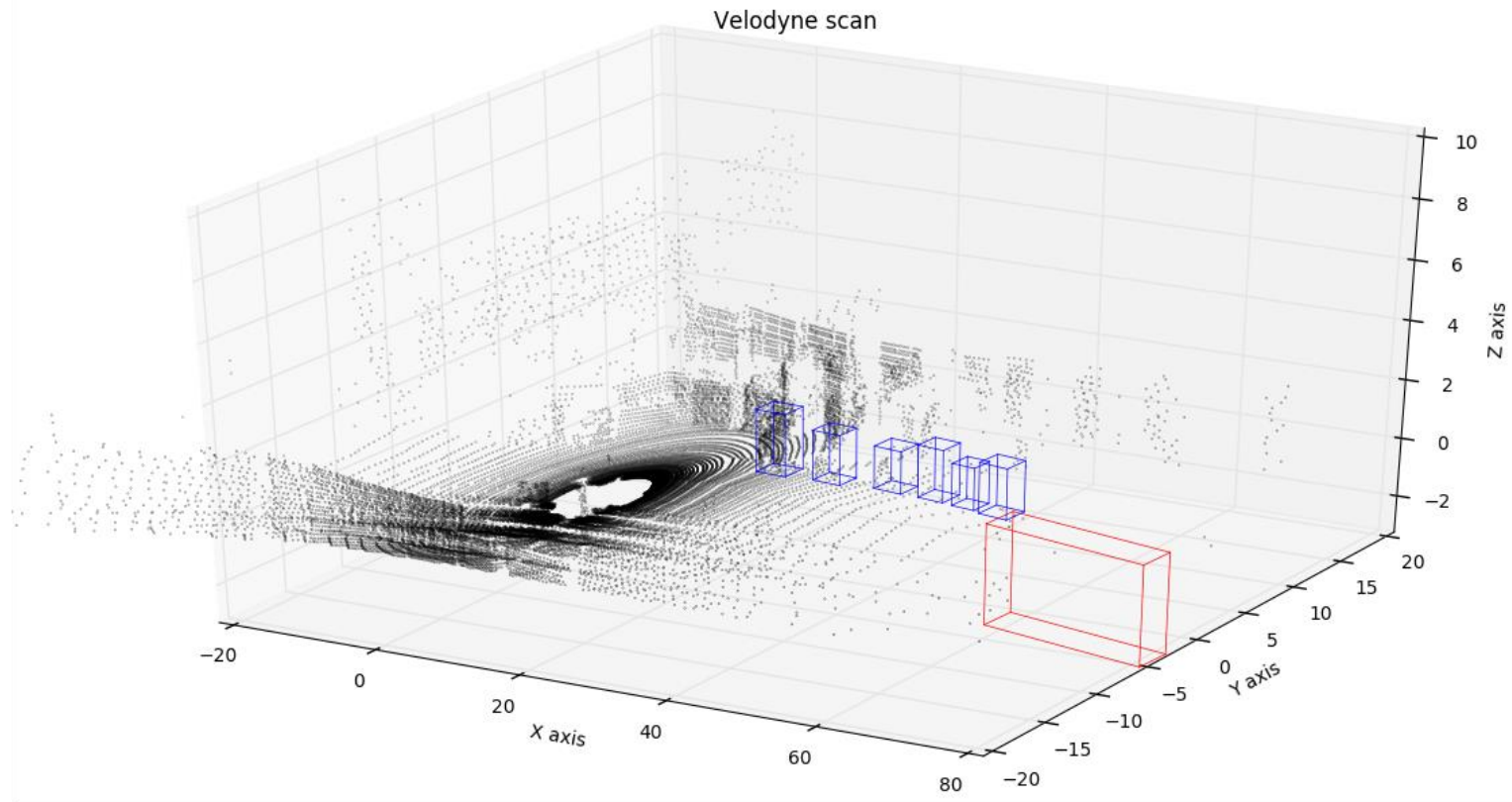
LIDAR



LIDAR

LiDAR sample frame 시각화

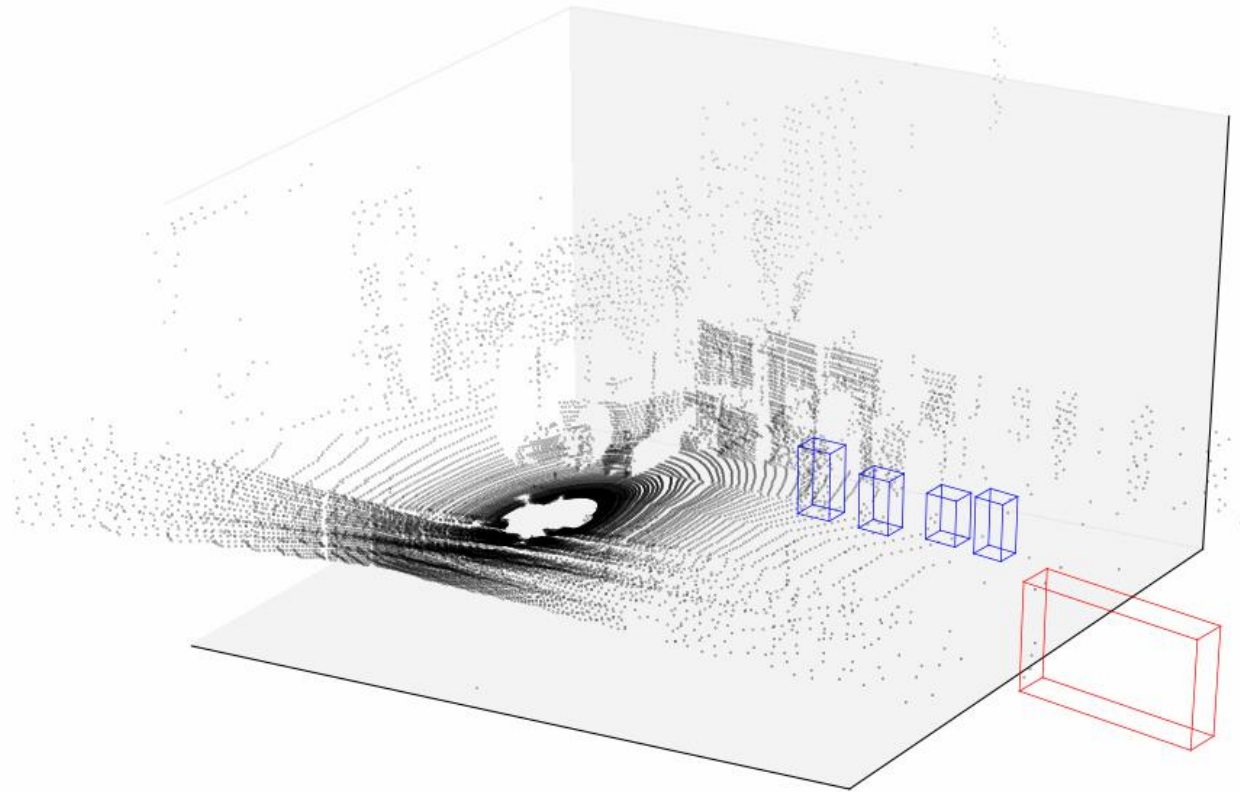
3D Point cloud



LIDAR

LiDAR frames 시각화

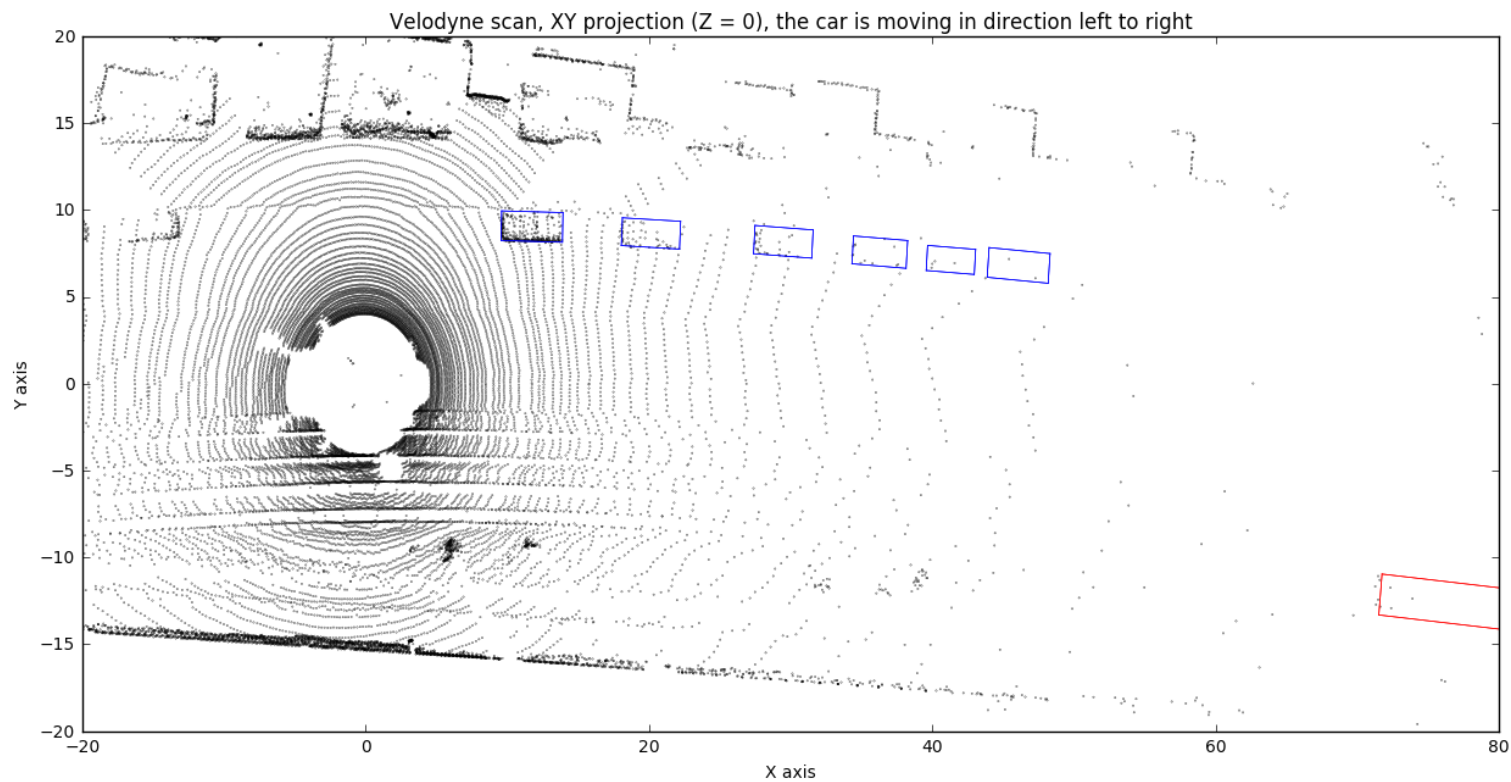
3D Point cloud



LIDAR

XY 축으로 투영 (Projection)

➔ 조감도 (Bird's-eye view, BEV)



LIDAR

장점

- 전범위 (360도) 감지 가능
- 깊이 (Depth) 정보 취득 가능
- 높은 해상도



LIDAR

장점

- 전범위 (360도) 감지 가능
- 깊이 (Depth) 정보 취득 가능
- 높은 해상도

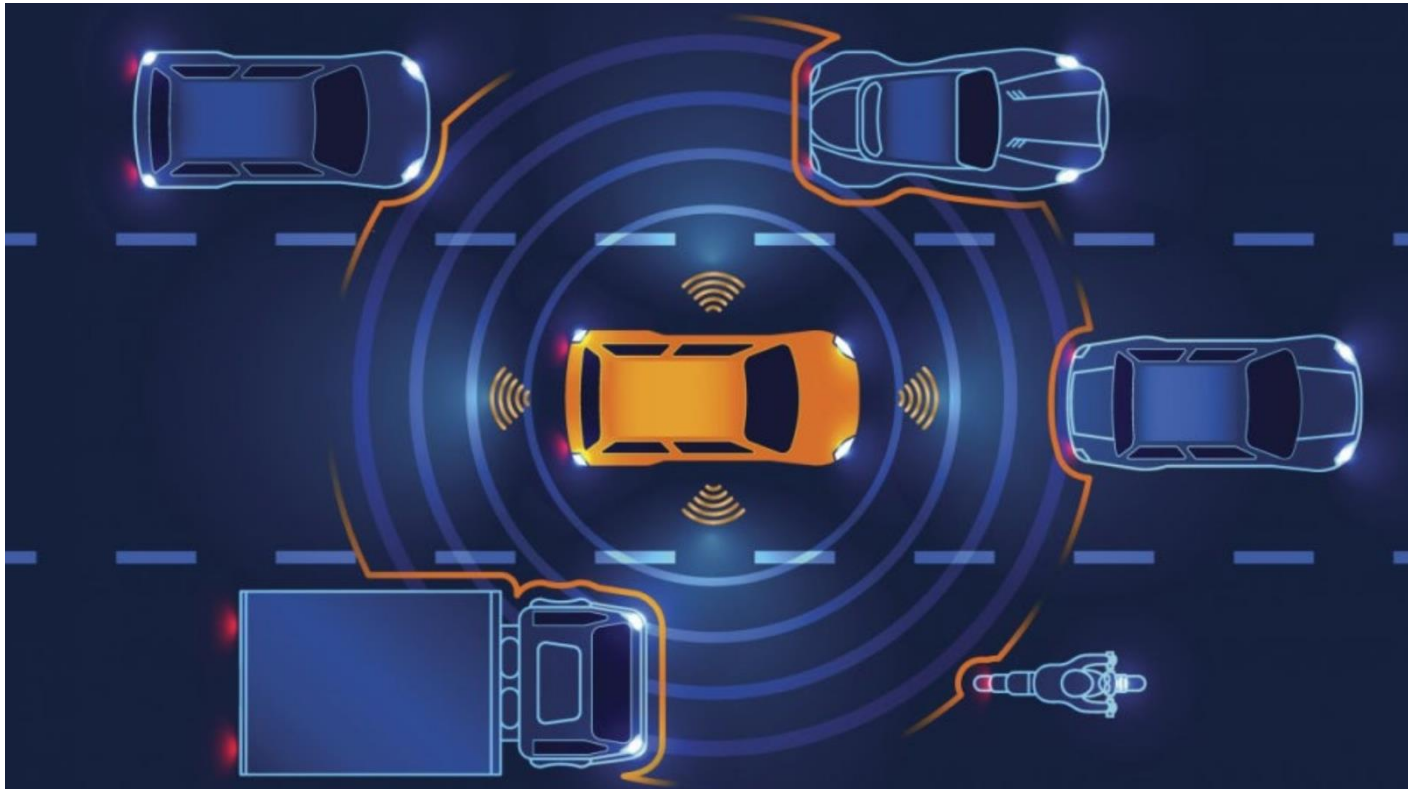
단점

- 색 인지 불가능
- 날씨 등 외부 환경에 영향
- 센서 가격이 매우 비쌈 → 점차 저렴해지고 있음
- 많은 전력 소모



RADAR

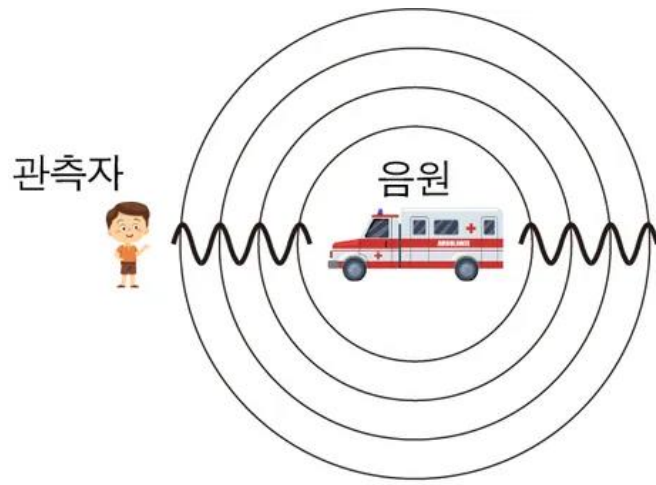
레이더(RADAR) : 전파 이용 탐지 및 거리 측정(RADio Detection And Ranging)



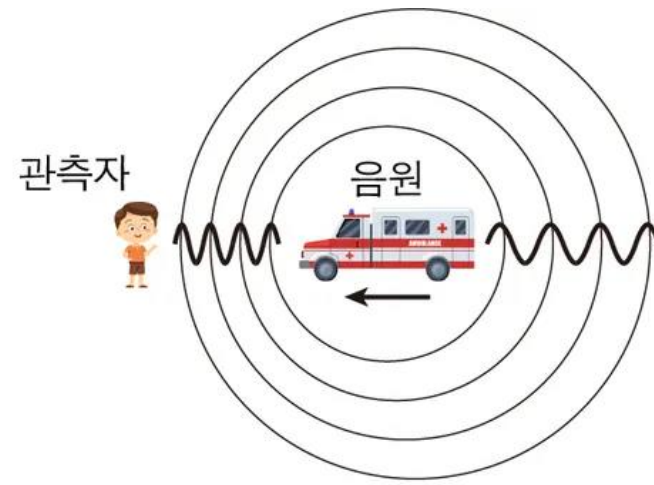
RADAR

장점

- 까다로운 기상 조건에도 물체 감지 가능
- 원거리 감지 가능
- 물체의 속도 감지 가능 (도플러 효과 활용)
- 가격이 상대적으로 저렴



(가) 음원이 정지했을 때



(나) 음원이 운동할 때

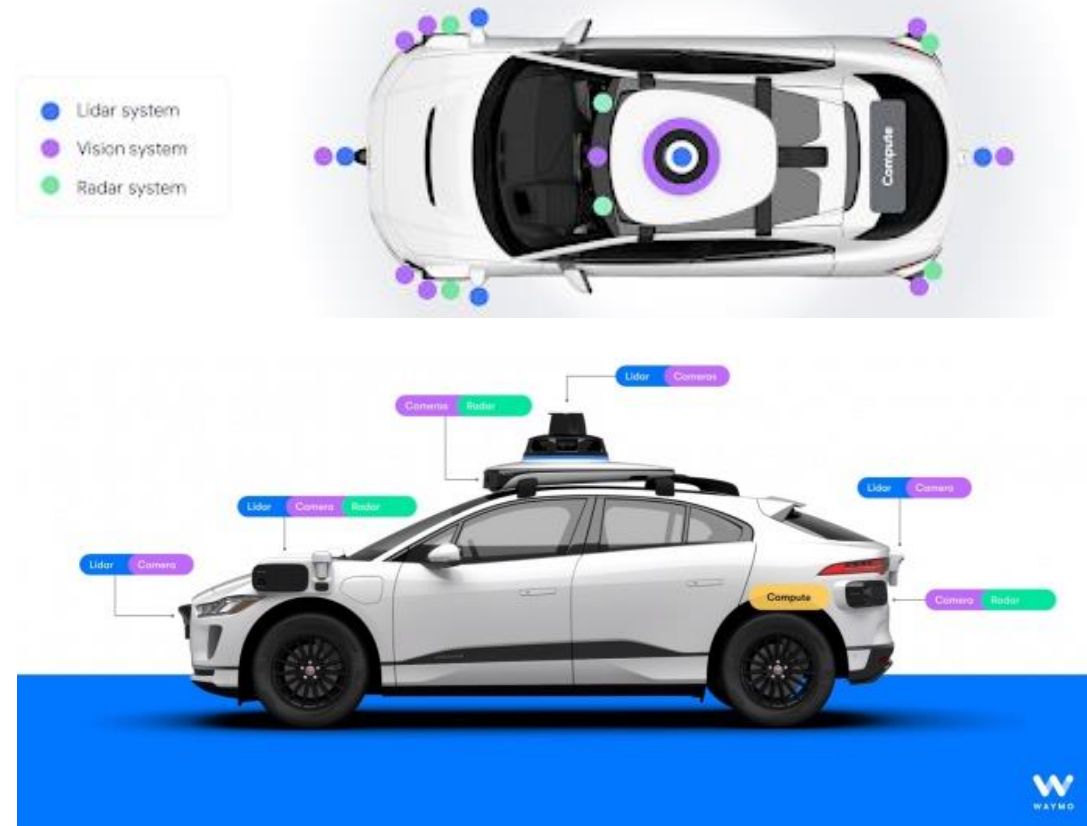
CASE STUDY

The image features a minimalist design on a light gray background. Two thin, dark gray lines intersect diagonally. One line runs from the top-left towards the bottom-right, while the other runs from the top-right towards the bottom-left. The text 'CASE STUDY' is positioned to the right of the intersection, rendered in a bold, black, sans-serif typeface.

EXAMPLES

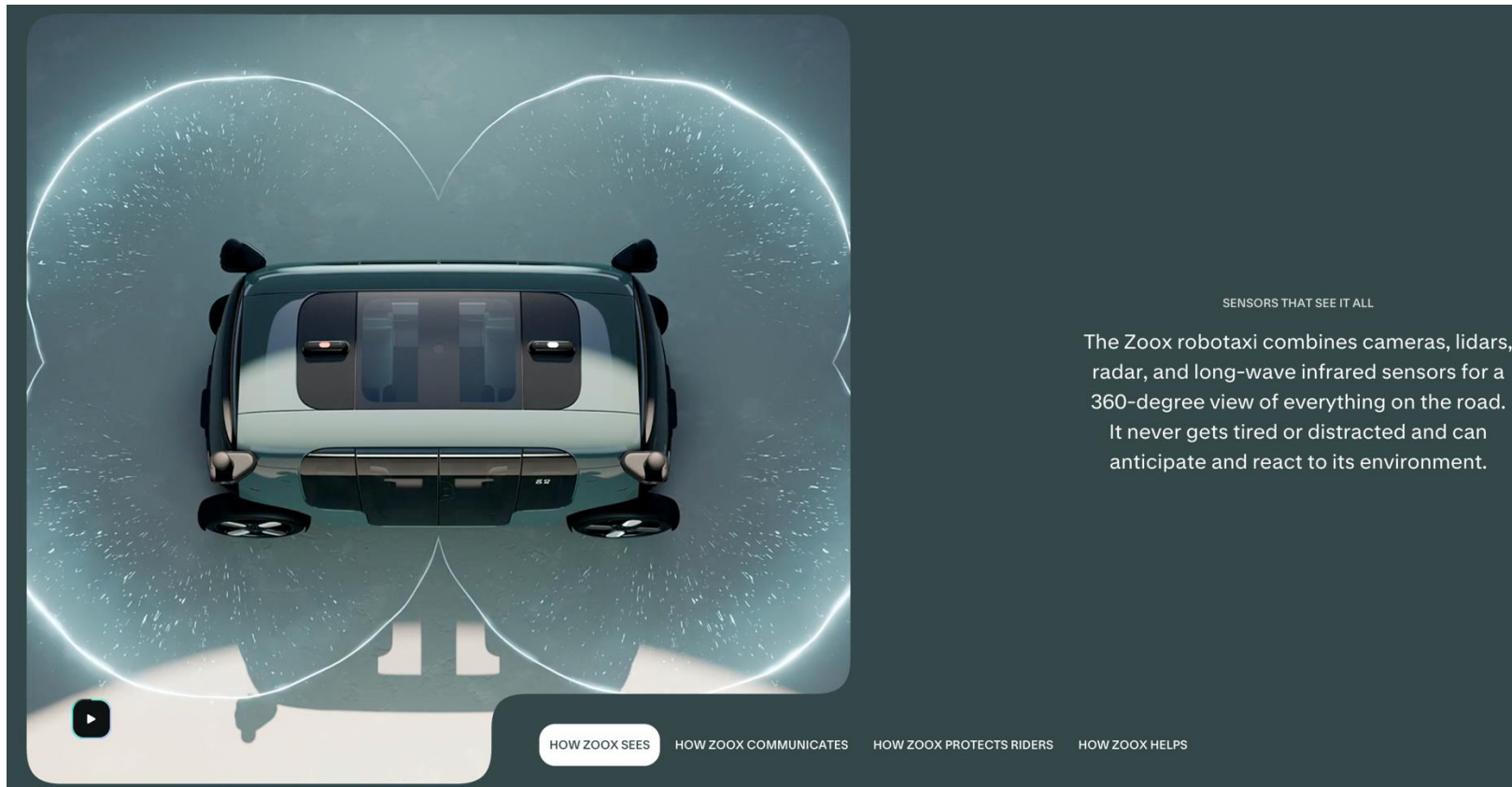
1) Waymo : 29 Cameras, 5 LiDARs, 6 Radars

The Waymo Driver



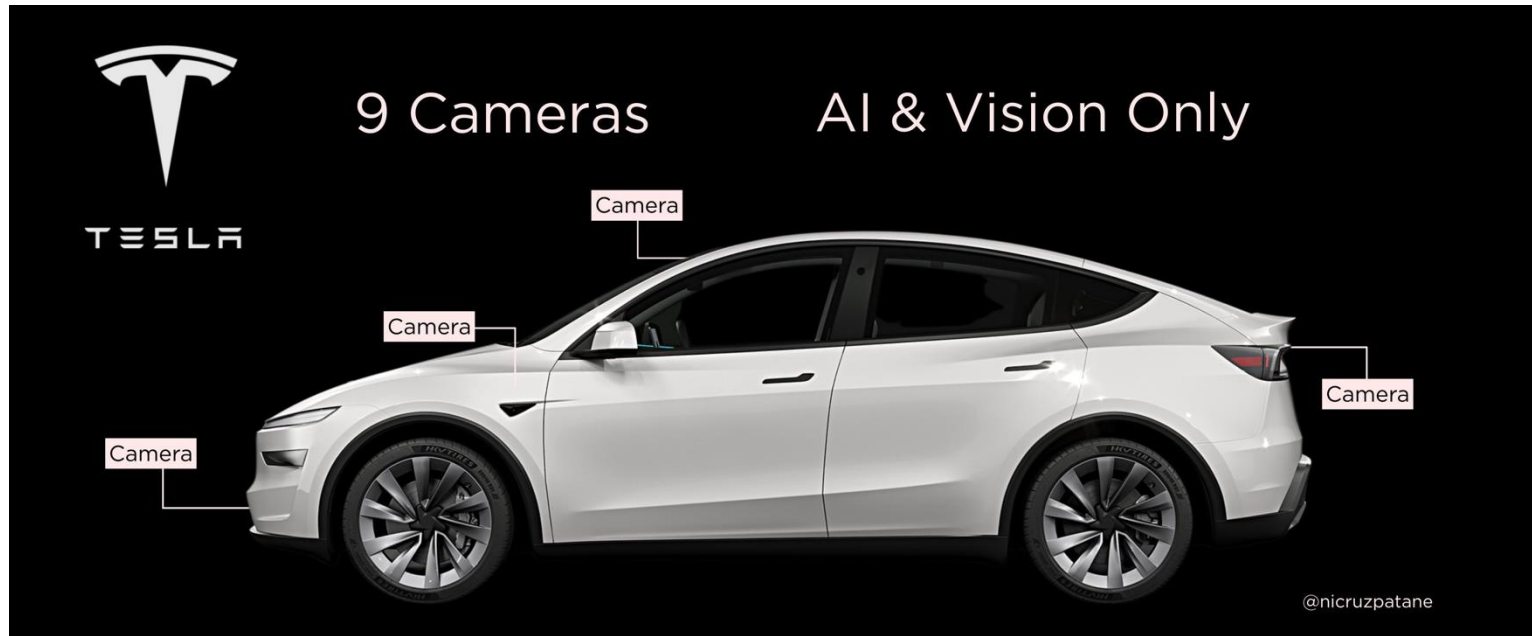
EXAMPLES

2) Zoox



EXAMPLES

3) Tesla



하지만...



An abstract graphic on a light gray background. Two thin, dark gray lines intersect. One line is nearly vertical, starting from the top center and extending towards the bottom right. The other line is nearly horizontal, starting from the top left and extending towards the right. The word "SUMMARY" is positioned to the right of the intersection point, aligned with the horizontal line.

SUMMARY

SUMMARY

- 1. Role of sensors in autonomous driving cars**
- 2. Types of sensors in autonomous driving cars**
 - Camera
 - LiDAR
 - RADAR
- 3. Case study**



NEXT LECTURE

Perception (1)



Two thin, dark gray lines intersect diagonally on a light gray background. One line slopes downwards from left to right, and the other slopes upwards from left to right.

QUESTIONS?